

1. Arrays, Subarrays, and Subsequences

Clarification: A **subarray** is contiguous; a **subsequence** maintains order but is not necessarily contiguous; a **subset** ignores order entirely.

Scenario	Formula	Example (n=3)
Total Subarrays (contiguous)	$\frac{n \cdot (n+1)}{2}$	6
Total Subsequences (non-empty)	$2^n - 1$	7
Total Subsequences (including empty)	2^n	8
Total Subsets (same as subsequences, order ignored)	2^n	8
Total subarrays of length exactly k	$n - k + 1$	(n=5,k=2) $\rightarrow$ 4

2. Permutations (Order Matters)

When arranging r items selected from n distinct items.

Scenario	Formula	Notes
Arrange all n items (no repetition)	$n!$	3 items $\rightarrow$ 6 ways
Arrange r items from n (no repetition)	$P(n, r) = \frac{n!}{(n-r)!}$	Order matters, no reuse
Arrange r items from n (with repetition)	n^r	PIN codes, passwords
Circular arrangements (distinct rotations)	$(n - 1)!$	Necklaces/round table (if reflections distinct)
Permutations with identical items	$\frac{n!}{k_1! \cdot k_2! \cdot \dots \cdot k_m!}$	Anagram of "MISSISSIPPI"
Derangements (no item in its place)	$!n = n! \sum_{i=0}^n \frac{(-1)^i}{i!}$	Nearest integer to $\frac{n!}{e}$

3. Combinations (Order Does Not Matter)

When selecting r items from n distinct items.

Scenario	Formula	Notes
Select r items (no repetition)	$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$	Standard binomial coefficient
Select r items (with repetition allowed)	$\binom{n+r-1}{r}$	Stars and Bars method
Select any number (0 to n)	2^n	Sum of all binomial coefficients
Select exactly half (if n even)	$\binom{n}{n/2}$	Largest binomial coefficient

4. Pairs and Tuples (Specific 2-element cases)

Extremely common in coding problems (Two-Sum, inversions, graph edges).

Scenario	Formula	Condition
Unordered pairs (e.g., $i < j$)	$\binom{n}{2} = \frac{n(n-1)}{2}$	No self-pairs, order ignored
Ordered pairs (e.g., $i \neq j$)	$n \cdot (n - 1)$	No self-pairs, order matters
Unordered pairs with repetition (e.g., $i \leq j$)	$\frac{n(n+1)}{2}$	Self-pairs allowed, order ignored
Ordered pairs with repetition (e.g., any i, j)	n^2	Self-pairs allowed, order matters
Number of coprime pairs in $[1..n]$	$\sim \frac{6}{\pi^2} n^2$	Asymptotic (Euler's totient sum)

5. Partitioning and Splitting

Dividing a set into groups.

Scenario	Formula	Notes
Partition n distinct items into exactly k non-empty, unlabeled subsets	$S(n, k)$ (Stirling numbers of the 2nd kind) $S(n, k) = \frac{1}{k!} \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n$	e.g., $S(4, 2) = 7$
Partition n distinct items into any number of subsets	Bell number $B_n = \sum_{k=0}^n S(n, k)$	$B(3) = 5$
Surjections (onto functions from n to k)	$k! \cdot S(n, k)$	Mapping n distinct balls into k distinct boxes, none empty
Integer Partitions (n identical items into k parts)	$p_k(n)$	No closed-form; uses generating functions

6. Graph and Tree Counts (CS Essentials)

Scenario	Formula	Notes
Total undirected graphs on n labeled vertices	$2^{\binom{n}{2}} = 2^{\frac{n(n-1)}{2}}$	Choose which edges exist
Total directed graphs on n labeled vertices	$2^{n(n-1)}$	Two possible directions per edge (no self-loops)
Number of labeled trees (Cayley's formula)	n^{n-2}	$n=4 \rightarrow 16$ trees
Number of undirected connected graphs (approx)	$2^{\binom{n}{2}} - \dots$	No simple closed form, use recurrence
Number of Binary Trees (unlabeled) / BSTs	$C_n = \frac{1}{n+1} \binom{2n}{n}$	n -th Catalan number
Number of Full Binary Trees (0 or 2 children)	C_{n-1}	where n = internal nodes
Number of ways to triangulate an $(n + 2)$ -gon	C_n	Catalan again!

7. Common Summations & Derived Counts

Useful for breaking down larger combinatorial problems.

Scenario	Formula
Sum of lengths of all subarrays	$\frac{n(n+1)(n+2)}{6}$
Sum of all subarray sums (if array is $[1, 2, \dots, n]$)	$\frac{n(n+1)^2(n+2)}{12}$
Number of bitstrings of length n with exactly k ones	$\binom{n}{k}$
Number of bitstrings with no consecutive 1s	F_{n+2} (Fibonacci sequence)
Inversions in a permutation (max possible)	$\frac{n(n-1)}{2}$
Total pairs (i, j) where $i < j$ and $a_i < a_j$	Depends on array, but max is $\frac{n(n-1)}{2}$ (strictly increasing)

Quick Cheat Sheet (Memory Aid)

Count what?	Formula (n = total, r = chosen)
Subarrays	$\frac{n(n+1)}{2}$
Subsets/Subsequences	2^n
Permutations (no rep)	$\frac{n!}{(n-r)!}$
Permutations (with rep)	n^r
Combinations (no rep)	$\binom{n}{r}$
Combinations (with rep)	$\binom{n+r-1}{r}$
Unordered Pairs ($i < j$)	$\binom{n}{2}$
Labelled Trees	n^{n-2}
Binary Trees	$C_n = \frac{1}{n+1} \binom{2n}{n}$